# Data Structures-I

## Session Objectives

- Understand the meaning of data structures and why we use them.
- Common data structures in Python
- Understand what lists are.
- Understand common methods and operations associated with lists.
- Understand the meaning of list comprehension
- Understand what tuples are.
- Understand common methods and operations associated with tuples.
- Understand the Comparison between Lists and Tuples

## Strings

```
# String Format .format() / f-string
name = "Paras"
age = 29
gender = "Male"

print("Hi, {}... You are {} year old. You have a Gender : {}.".format(name,age,gender))
```

```
Hi, Paras... You are 29 year old. You have a Gender : Male.
```

```
print(f"Hi, {name}... You are {age} year old. You have a Gender : {gender}.")
```

```
Hi, Paras... You are 29 year old. You have a Gender : Male.
```

```
# Strip() -> it removes leading and trailing whitespace [also can remove special characters]
print("          Python Programming          ".strip()) # default = " "
```

```
Python Programming
```

```
print("**********Python Programming**********".strip("*"))
```

```
Python Programming
```

```
print("$$$$$$$$$$Python Programming**********".strip("*"))
```

```
$$$$$$$$$$Python Programming
```

```
print("$$$$$$$$$$Python Programming**********".strip("*").strip("$"))
```

```
Python Programming
```

```python
# .index() -> returns the position of a string
_str = "This is a Python Course!"
print(_str.index('is')) # 2
```

```
2
```

```python
_str = "This is a Python Course!"
print(_str.index("Python")) # 10
```

```
10
```

```
# Python is a case-sensitive language
_str = "This is a Python Course!"
print(_str.index("python")) # ValueError: substring not found
```

```python
_str = "This is a Python Course!"
print(_str.index('is',3)) # 5
```

```
5
```

```python
print(_str.index('is',3,10))
```

```
5
```

```
# index(substr,start=0,stop=len(Non-Inclusive))
```

```python
# String Check() -> Validator [Boolean Returns]
print("CondingNinjas".isalpha()) # True
print("CondingNinjas123".isalpha()) # False
print("Ninjas123".isalnum()) # True
print("PythonProgram".isupper()) # False
print("PYTHON".isupper()) # True
print("python".islower()) # True
print("31494369631".isnumeric()) # True
print("python631".isnumeric()) # False
print("CodingNinjas9911".swapcase()) # Convert uppercase -> Lowercase and vice-verse
```

```
True
False
True
False
True
True
True
False
cODINGnINJAS9911
```

```python
print("coding123".islower()) # True
```

```
True
```

```python
# startswith() -> Boolean Returns
_str = "Hey, Welcome to the World of Programming!"
print(_str.startswith("Hey")) # True
print(_str.startswith("hey")) # False
print(_str.startswith("Hello")) # False
print(_str.startswith("H")) # True
print(_str.startswith("Hi")) # False
print(_str.startswith("Welcome",5)) # True
```

```
True
False
False
True
False
True
```

```python
# endswith() -> Boolean Returns
_str = "Hey, Welcome to the World of Programming!"
print(_str.endswith("hey")) # False
print(_str.endswith("Program")) # False
print(_str.endswith("Programming!")) # True
print(_str.endswith("!")) # True
print(_str.endswith("g!")) # True
print(_str.endswith("ing!",-4)) # True
print(_str.startswith("ing!",-4)) # True
print(_str.endswith("ing!",-3)) # False
```

```
False
False
True
True
True
True
True
False
```

```python
# .count() -> 'substring' -> count the occurrence of substring
_str = "This is a Python Course!"
print(_str.count('is')) # 2
```

```
2
```

```python
_str = "This is a Python Course!"
print(_str.count('s')) # 3
```

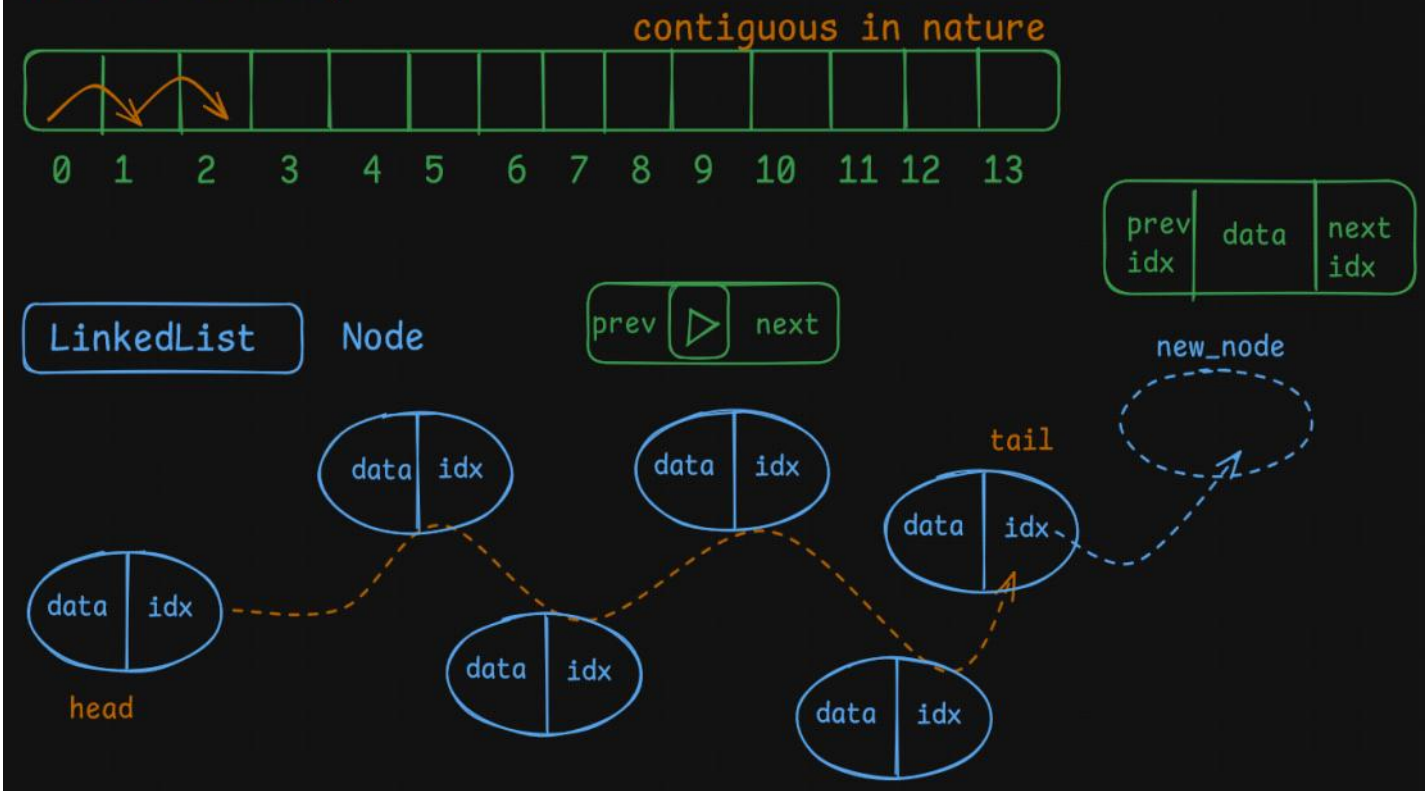
```
3
```

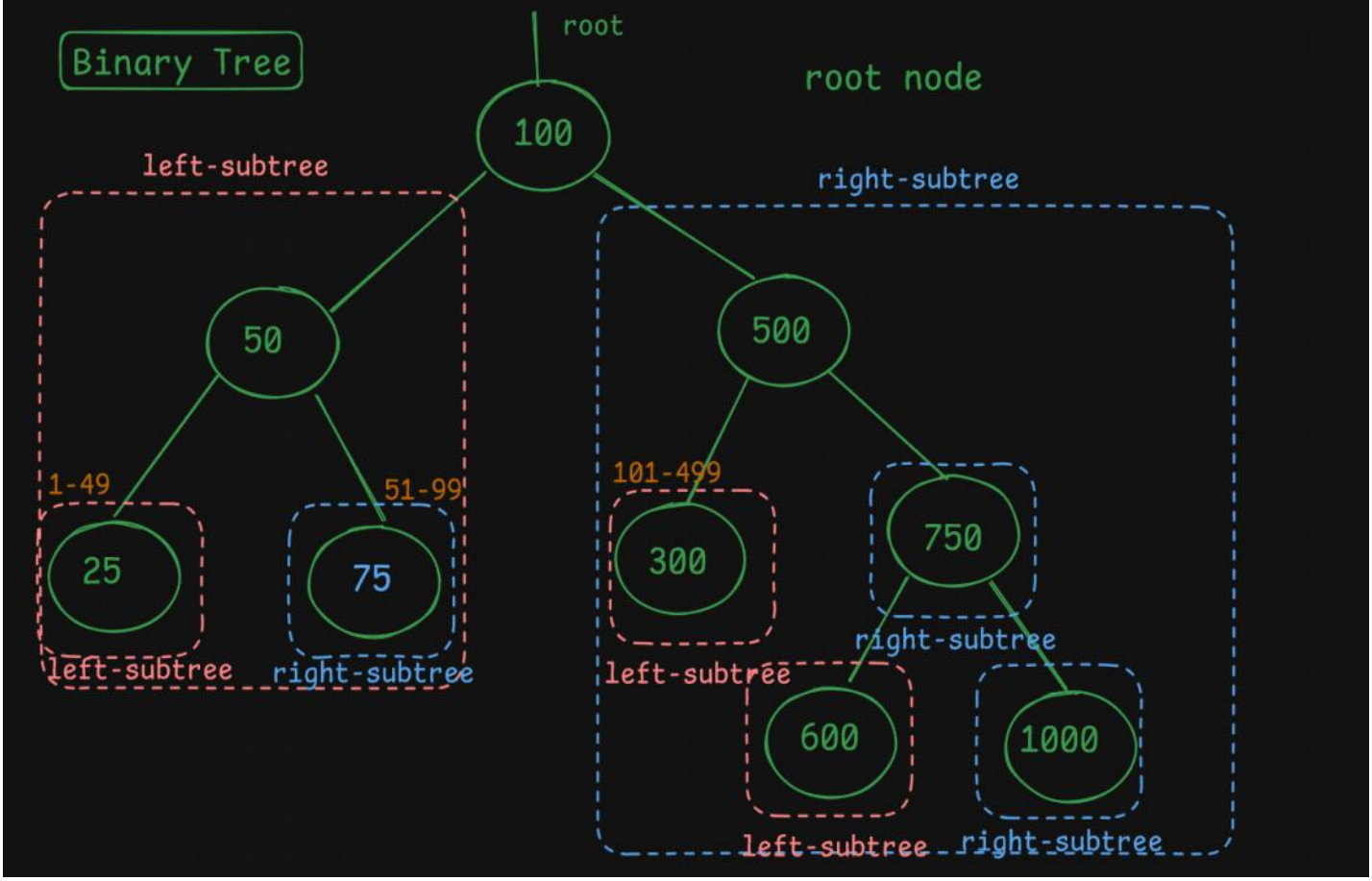
```python
# Membership Operators 'in'
print('is' in _str) # True
```

```
True
```

Data Structure | List | Array + LinkedList

What is an Array:



# What is a Data Structure?

A data structure is simply a way to organize and store data in memory, so that we can use it efficiently.

## Why do we need to use them?

1. Organizations : Keep the data neat and easy to find.
2. Efficiency - Helps to perform operations (insert, delete, search) quickly.
3. Memory Management - Savas space, avoid waste.
4. Flexibility - Can store all sorts of data (number, text, etc.)
5. Abstraction - Hides the complex details, we just call simple operations
6. Scalability - Good Data Structure lets our program handle more data without slowing down.

# Lists in Python:

A List is :

1. An ordered collection (items have a fixed position).
2. mutable (can be changed after creation).
3. Allows Duplicates.
4. Lists are also flexible -> they can store different data types together like integers, string, booleans, etc.

## Why use lists?

- To keep a group of related items togethers in a specific order.
- To easily add, remove, change, or access the elements.
- To perform the task like sorting, or summing values.

```
_list = []
print(_list) # []
print(type(_list)) # 'list'
```

```
[]
<class 'list'>
```

```
_list = [1,2,3,4,5] # Similar D-types [Integer List]
print(_list)
```

```
[1, 2, 3, 4, 5]
```

```
_list = [1,2.9,False,"Python",'k'] # Mixed Dtypes
print(_list)
```

```
[1, 2.9, False, 'Python', 'k']
```

```
# Duplicate Elements are allowed in a list
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']
print(country_list)
print(type(country_list))
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China']
<class 'list'>
```

```python
# List Constructor()
country_list = ('India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China')
print(country_list)
print(type(country_list)) # 'tuple' () # immuatable
```

```
('India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China')
<class 'tuple'>
```

```python
# List Constructor()
country_list = list(('India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China'))
print(country_list)
print(type(country_list)) # 'list' [] # mutable
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China']
<class 'list'>
```

```python
_list = list("Coding") # string
print(_list) # ['C','o','d','i','n','g']
print(type(_list)) # 'list'
```

```
['C', 'o', 'd', 'i', 'n', 'g']
<class 'list'>
```

```python
_list = list(("Coding",)) # tuple
print(_list) # ["Coding",]
print(type(_list)) # 'list'
```

```
['Coding']
<class 'list'>
```

```python
_list = list(["Coding"]) # list
print(_list) # ["Coding"]
print(type(_list)) # 'list'
```

```
['Coding']
<class 'list'>
```

```python
val = ("Python")
print(val)
print(type(val)) # 'str'
```

```
Python
<class 'str'>
```

```python
val = ("Python",)
print(val)
print(type(val)) # 'tuple'
```

```
('Python',)
<class 'tuple'>
```

```python
# Nested List : List inside List is a nested List..... [2 Dimension]
nested_list = [
    111,
    'Coding',
    [11,22,33,44,55],
    'Python',
    ['a','b','c','d','e'],
    True
]
print(nested_list)
```

```
[111, 'Coding', [11, 22, 33, 44, 55], 'Python', ['a', 'b', 'c', 'd', 'e'], True]
```

```python
print(nested_list[0]) # 111
print(nested_list[2]) # [11, 22, 33, 44, 55]
print(nested_list[2][-1]) # 55
print(nested_list[-2][-5]) # 'a'
print(nested_list[-4][2]) # 33
# print(nested_list[4][5]) # Error
```

```
111
[11, 22, 33, 44, 55]
55
a
33
```

```python
# Indexing
# Slicing [Start:0, Stop:Len(Non-Inclusive), Step:+1(default)]
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']

print(country_list[0]) # 'India'
print(country_list[5]) # 'SA'
print(country_list[-4]) # 'India'
print(country_list[-9]) # 'Nepal'
print(country_list[14]) # 'Canada'
print(country_list[-17]) # 'Ukraine'
```

```
India
South Africa
India
Nepal
Canada
Ukraine
```

```python
# Slicing [Start:0, Stop:Len(Non-Inclusive), Step:+1(default)]
# Sub-List
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']

print(country_list[::]) # Full List
print(country_list[2:5]) # ['Ukraine','New Zealand','Australia']
print(country_list[9:]) # ['Germany', ..... , 'China']
print(country_list[-5:]) # ['Canada' To 'China'] # Step : +1
print(country_list[-9:2]) # []
print(country_list[-9:2:-1]) # ['Nepal' to 'NZ']
print(country_list[-4:-2]) # ['India','America']
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China']
['Ukraine', 'New Zealand', 'Australia']
['Germany', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China']
['Canada', 'India', 'America', 'Germany', 'China']
[]
['Nepal', 'Germany', 'Spain', 'Italy', 'Russia', 'South Africa', 'Australia', 'New Zealand']
['India', 'America']
```

```python
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']

print(country_list[-7:1:2]) # []
print(country_list[-7:1:-2]) # [iran->nepal->spain->russia->aus->ukraine]
print(country_list[7::3]) # [italy >> nepal >> pakista >> america]
print(country_list[-9:-2:-1]) # []
print(country_list[-9:-2:2]) # ['Nepal','Iran','Canada','America']
print(country_list[-17:15:4]) # [ukraine->russia->nepal->canada]
```

```
[]
['Iran', 'Nepal', 'Spain', 'Russia', 'Australia', 'Ukraine']
['Italy', 'Nepal', 'Pakistan', 'America']
[]
['Nepal', 'Iran', 'Canada', 'America']
['Ukraine', 'Russia', 'Nepal', 'Canada']
```